COAL CONCEPTS

SEMI-PREPARED / TOTAL MOISTURE/ ROGA INDEX & FREE SWELLING INDEX PROFICIENCY TESTING

REPORT SEVENTY-ONE

Revision: 00

Final report

DATE ISSUED 28th FEBRUARY 2026

LABORATORY CODE: 143e

R BABOOLAL (SCHEME MANAGER)

S MAMABOLO (SCHEME COORDINATOR)

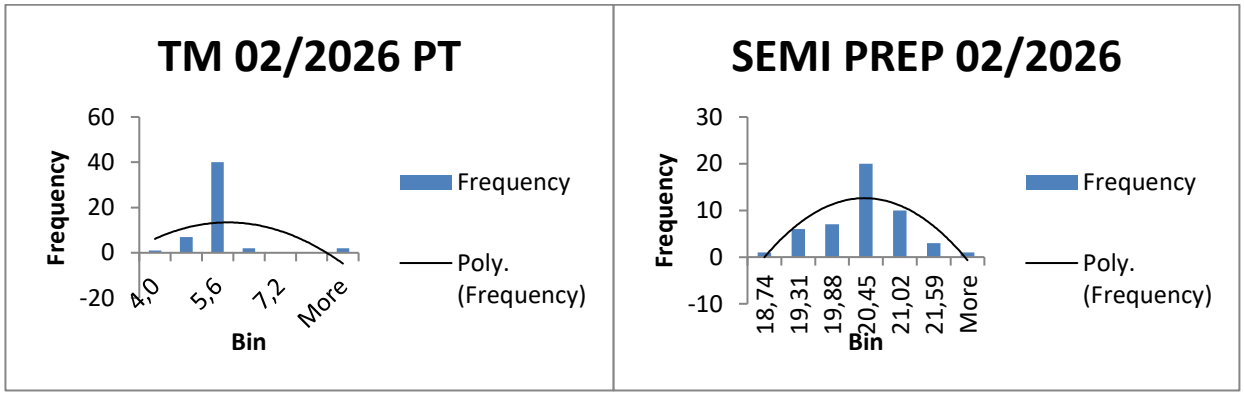
EXECUTIVE SUMMARY

1. Fifty-Seven samples were sent to participants with 56 results submissions.

2. Due to insufficient data for Roga Index and CSI statistical analysis could not be carried out.

3. Three outliers were detected, using Grubbs estimate for Total Moisture. Robust statistics were applied to calculate the robust standard deviation and robust average; the distribution of results did follow a Gaussian curve (below).

4. One outlier was detected for sample preparation with Ash determination as the indicator. Robust statistics were applied to calculate the robust standard deviation and robust average, the distribution of results did follow a Gaussian curve (below)



5. The trending of z-scores over time is a good indication of the laboratory's performance.

Report No	TM	ISO Ash
69	0,12	-1,32
71	-1.13	0.39

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Dear Participant

RE: SEMI-PREPARED SAMPLE PROFICIENCY TESTING RESULTS FOR THE MONTH OF FEBRUARY 2026

Thank you for your participation in the Coal Concepts proficiency testing scheme.

Your laboratory code is **as per the cover page.**

All results are totally confidential. Any results in bold, italics and underlined are outliers. Where applicable, the most extreme outliers have been eliminated from calculation of averages using the Grubbs estimate for outliers. Please take note of the following:

1. Z-scores between -1 and +1 is deemed acceptable

2. Z-scores between -2 and -3 should serve as a warning that the analysis result could get worse

3. Z-scores between +2 and +3 should also serve as a warning that analysis results could get worse.

4. Z- scores lower than -3 and exceeding +3 should warrant an investigation

6. All calculations can be made available upon request

The Coal Concepts scheme adheres to the requirements of ISO/IEC 17043 – Conformity assessment – General requirements for proficiency testing.

Please find results attached together with Z-score trends.

Best Regards

R Baboolal

LIST OF PARTICIPANTS IN ALPHABETICAL ORDER

Africoal Lab
Alfred H Knight
Ağrı İbrahim Çeçen Üniversitesi
Azuresky Pty LTd
Best Enough Laboratory-Springs
Best Enough Laboratory-2 Seam
Bestech Vlakfontein Mine-Ogies
Bureau Veritas Inspectorate Laboratories Alton
Bureau Veritas-Dubai
Bureau Veritas-Matola
Bureau Veritas Inspectorate Laboratories Middelburg
Bureau Veritas Inspectorate Laboratories Tendele
Bureau Veritas Moatize Laboratory
Bureau Veritas Nacala Laboratory
CCIC Africa Richards Bay
Cigroup ACE
Cigroup Africa
Coal Connect Lab Services
Cotecna Phola
Cotecna South Africa Middelburg
Cotecna South Africa Richards Bay
ELI Turkey
Eskom Holdings - Arnot Power Station
Eskom Holdings - Duvha Power Station
Eskom Holdings - Grootvlei Power Station
Eskom Holdings - Hendrina Power Station
Eskom Holdings - Kendal Power Station
Eskom Holdings - Kriel Power Station
Eskom Holdings - Lethabo Power Station
Eskom Holdings - Majuba Power Station
Eskom Holdings – Matla Power Station
Eskom Holdings – Tutuka Power Station
Exxaro Matla
Eyethu Coal Wilge
Fauji Fertilizer Bin Qasim Limited
G&W Base and Industrial Minerals
Geoscience
GLI Turkey
Highveld Laboratory Services
Imbally (Pty) Ltd
Imbally (Pty) Ltd-Mooiplaats
Jindal Mozambique
KOTEKNA GÖZETİM A. Ş
KOREA WESTERN POWER CO., Ltd.
Labrite Ptv Ltd
Leon Inspection and Testing - Pakistan
MITRA SK MOROCCO SARL
Mitra SK - Richards Bay
Mpumamanzi
Noko Analytical Services-Witbank
Noko Analytical Piet Retief
RC Inspection
PJS Middelburg
Polyak Eynez Coal Laboratory: Turkey
PSB Analytical Services Netherlands
Ronewa Lab
Ronewa Gugulethu
Ruvuma Coal LTD
SA Labs Ithuba Khanye
SA Labs Ithuba Middelburg
SA Labs Ithuba ZAC
SA Labs Ithuba-ukufisa
SABS Newcastle
SABS Uitkomst
SB Mining Belfast
Seriti Power - VDD
SGS Italia SpA
Sibonisiwe Middelburg
Siza Coal Service Botswana
Siza Coal Service Dundee
Siza Coal Service Leeuwpan
Siza Coal Services Kinross
Siza Coal Services Middelburg
Siza Coal Services NBC
Siza Coal Services Wes Coal
Siza Coal Services Wildfontein
South 32 Khutala
Turkey Coal-Department of garp lignites operation
Turkish Coal Enterprises General Directorate
Turkish GLI
Umzamo Analytical Services-Botswana
Umzamo Analytical Services – Hendrina
Umzamo Analytical Services – Main Lab
Umzamo Analytical Services - SCS
Umzamo Analytical Services-Twistdraai
Umzamo Analytical Services - Witbank
Vitrovian Analytical Services
Yatagan Termik
Yıldız Laboratuvar
Ykenerji

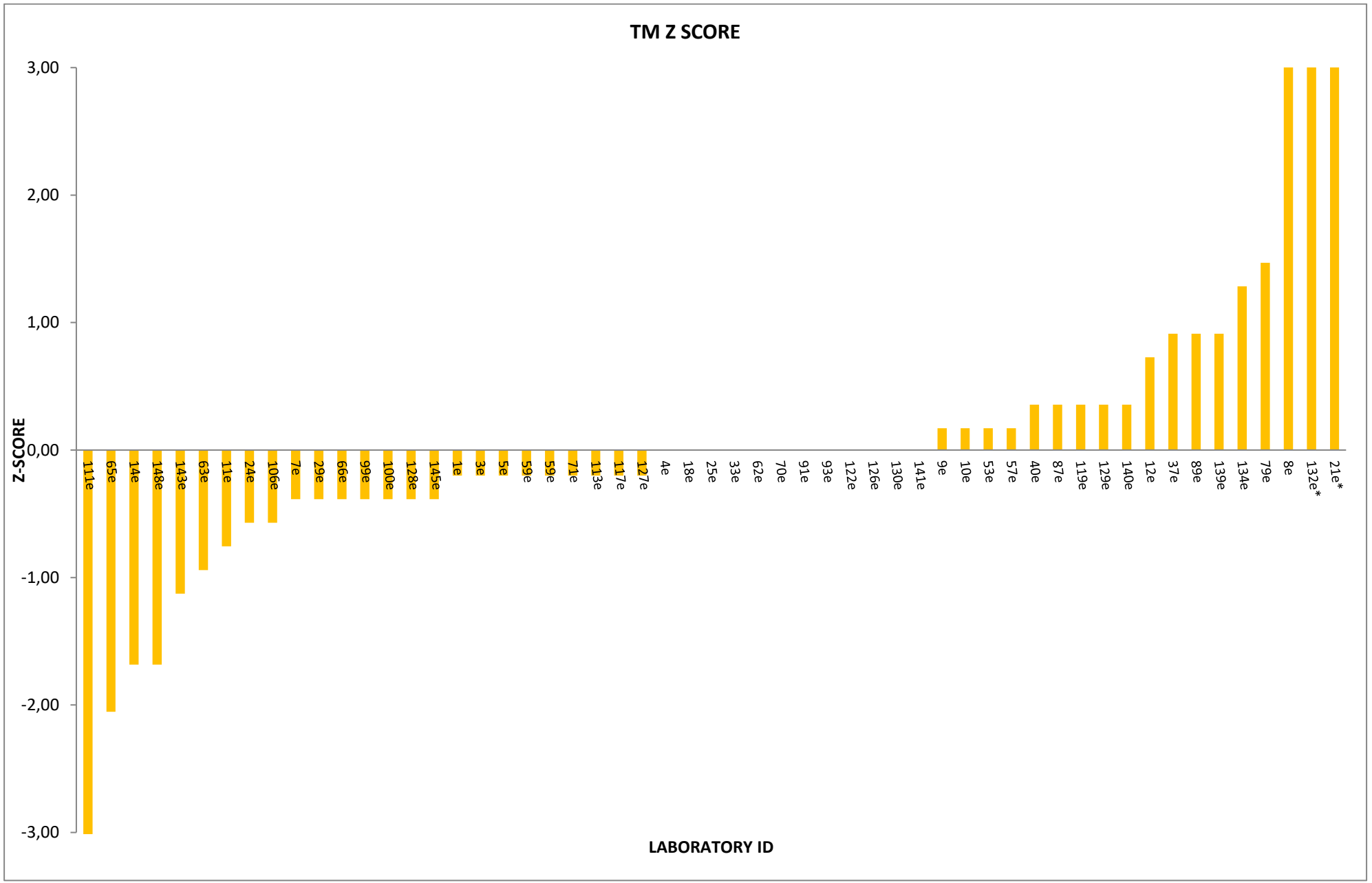
1. **TYPE OF SAMPLE USED**

   The coal used in this proficiency testing round was bituminous coal from the Mpumalanga region.

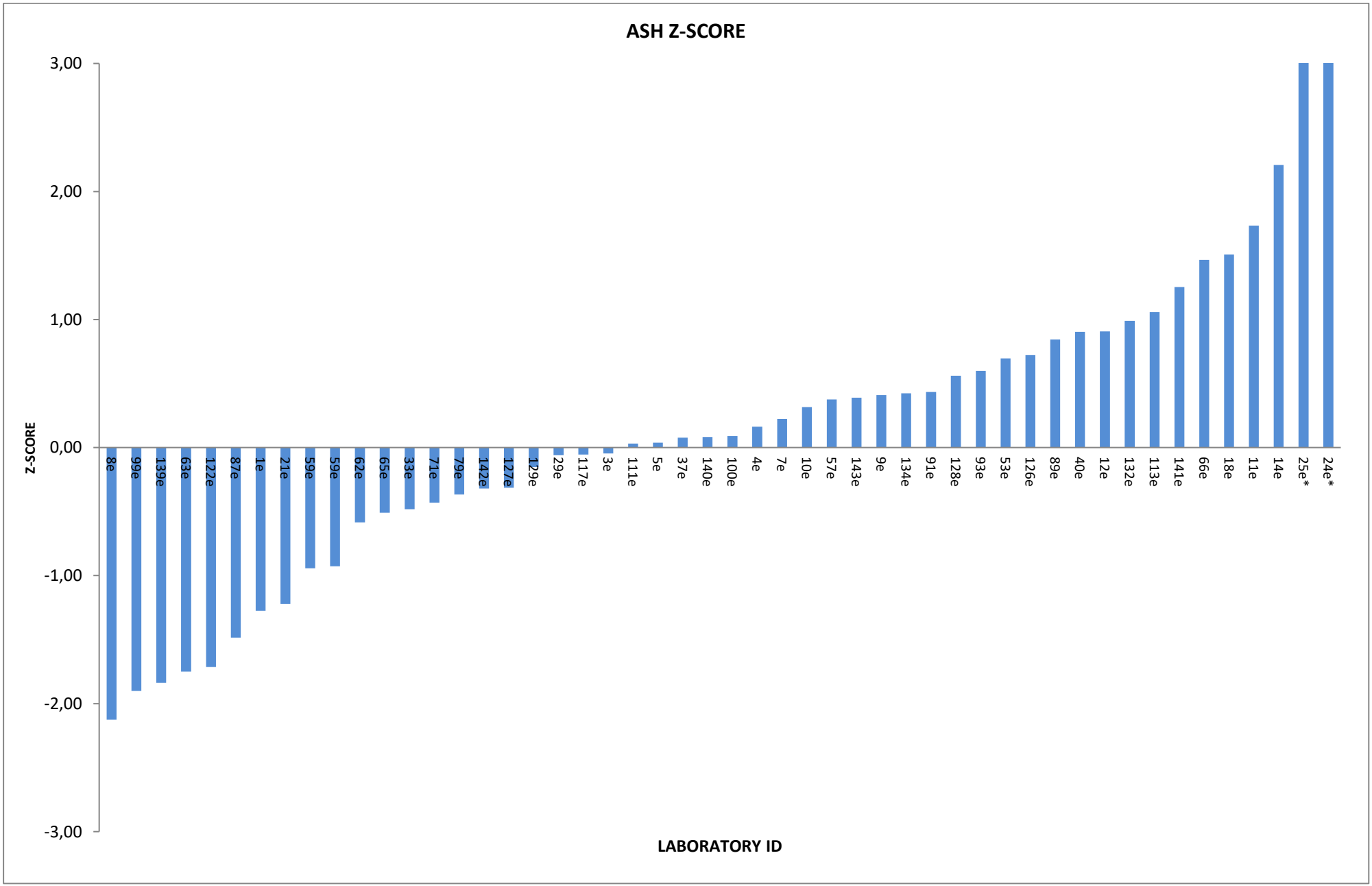
2. PREPARATION OF SAMPLE

Approximately 180kg's of sample with an approximate topsize of 50mm was sourced. This was crushed to -4.75 mm using a jaw crusher. The 4.75mm screen was placed on a 600um screen and the -4.75 mm material screened in batches of about 5kgs. Coal passing through the 4.75mm screen but retained on the 600um screen was placed in a mixing drum. Once all the coal was screened and transferred to the mixing drum, it was mixed for approximately 4 hours. The material was then transferred to containers capable of holding about 2kg of coal sample. Fifty-five samples were obtained in this way.

COAL CONCEPTS - PROFICIENCY TESTING - FEBRUARY 2026			
ANALYTICAL PARAMETER: TOTAL MOISTURE (%)			
	LAB ID	**AS RECEIVED (%)**	**Z-SCORE (AR)**
	1e	5,0	-0,20
	3e	5,0	-0,20
	4e	5,1	0,00
	5e	5,0	-0,20
	7e	4,9	-0,39
	8e	8,0	5,36
	9e	5,2	0,17
	10e	5,2	0,17
	11e	4,7	-0,76
	12e	5,5	0,73
	14e	4,2	-1,68
	18e	5,1	0,00
	21e*	***11,5***	***11,85***
	24e	4,8	-0,57
	25e	5,1	0,00
	29e	4,9	-0,39
	33e	5,1	0,00
	37e	5,6	0,91
	40e	5,3	0,36
	53e	5,2	0,17
	57e	5,2	0,17
	59e	5,0	-0,20
	59e	5,0	-0,20
	62e	5,1	0,00
	63e	4,6	-0,94
	65e	4,0	-2,05
	66e	4,9	-0,39
	70e	5,1	0,00
	71e	5,0	-0,20
	79e	5,9	1,47
	87e	5,3	0,36
	89e	5,6	0,91
	91e	5,1	0,00
	93e	5,1	0,00
	99e	4,9	-0,39
	100e	4,9	-0,39
	106e	4,8	-0,57
	111e	***3,2***	***-3,54***
	113e	5,0	-0,20
	117e	5,0	-0,20
	119e	5,3	0,36
	122e	5,1	0,00
	126e	5,1	0,00
	127e	5,0	-0,20
	128e	4,9	-0,39
	129e	5,3	0,36
	130e	5,1	0,00
	132e*	***9,5***	***8,14***
	134e	5,8	1,28
	139e	5,6	0,91
	140e	5,3	0,36
	141e	5,1	0,00
	143e	4,5	-1,13
	145e	4,9	-0,39
	148e	4,2	-1,68
Number of results	-	**55**	-
OUTLIERS	-	**3**	-
AVERAGE	-	**5,1**	-
STD DEVIATION	-	**0,5**	-
MEDIAN		**5,1**	
MAXIMUM		**8,0**	
MINIMUM		**4,0**	
RANGE		**4,0**	
%RSD	-	**10,6**	
ROBUST AVERAGE	-	**5,1**	-
ROBUST STD DEVIATION	-	**0,4**	-
UoM		**0,1**	
Upper reproducibility limit		**6,6**	
Lower reproducibilty limit		**3,6**	


TM Z SCORE
3,00
2,00
1,00
0,00
-1,00
-2,00
-3,00
Z-SCORE
111e
65e
14e
148e
143e
63e
11e
24e
106e
7e
29e
66e
99e
100e
128e
145e
1e
3e
5e
59e
59e
71e
113e
117e
127e
4e
18e
25e
33e
62e
70e
91e
93e
122e
126e
130e
141e
9e
10e
53e
57e
40e
87e
119e
129e
140e
12e
37e
89e
139e
134e
79e
8e
132e*
21e*
LABORATORY ID

COAL CONCEPTS - PROFICIENCY TESTING - FEBRUARY 2026					
ANALYTICAL PARAMETER: ISO ASH SEMI PREP PT (%)					
	LAB ID	MOISTURE IN ANALYSIS SAMPLE (%)	AIR DRY	DRY BASE	Z-SCORE (DRY BASE)
	1e	2,32	18,85	19,30	-1,28
	3e	2,73	19,56	20,11	-0,05
	4e	2,01	19,84	20,25	0,16
	5e	2,80	19,60	20,16	0,04
	7e	2,40	19,80	20,29	0,22
	8e	1,80	18,40	18,74	-2,13
	9e	2,50	19,90	20,41	0,41
	10e	2,20	19,90	20,35	0,32
	11e	2,93	20,66	21,28	1,73
	12e	2,21	20,28	20,74	0,91
	14e	2,57	21,04	21,59	2,21
	18e	3,00	20,50	21,13	1,51
	21e	2,45	18,86	19,33	-1,22
	24e*	2,13	***23,42***	***23,93***	***5,75***
	25e*	2,75	***22,14***	***22,77***	***3,98***
	29e	2,34	19,63	20,10	-0,06
	33e	2,23	19,38	19,82	-0,48
	37e	1,59	19,87	20,19	0,08
	40e	2,10	20,30	20,74	0,90
	53e	3,20	19,94	20,60	0,70
	57e	1,90	20,00	20,39	0,38
	59e	2,65	19,00	19,52	-0,94
	59e	2,70	19,00	19,53	-0,93
	62e	2,30	19,30	19,75	-0,58
	63e	2,13	18,58	18,98	-1,75
	65e	3,40	19,13	19,80	-0,51
	66e	2,40	20,60	21,11	1,47
	71e	2,80	19,30	19,86	-0,43
	79e	2,00	19,50	19,90	-0,37
	87e	2,40	18,70	19,16	-1,48
	89e	2,40	20,20	20,70	0,84
	91e	2,28	19,96	20,43	0,43
	93e	2,60	20,00	20,53	0,60
	99e	3,10	18,30	18,89	-1,90
	100e	2,02	19,79	20,20	0,09
	111e	2,83	19,59	20,16	0,03
	113e	2,10	20,40	20,84	1,06
	117e	2,85	19,53	20,10	-0,06
	122e	3,20	18,40	19,01	-1,71
	126e	3,47	19,90	20,62	0,72
	127e	1,97	19,54	19,93	-0,31
	128e	1,56	20,19	20,51	0,56
	129e	2,44	19,55	20,04	-0,15
	132e	2,37	20,30	20,79	0,99
	134e	2,10	19,99	20,42	0,42
	139e	1,20	18,70	18,93	-1,84
	140e	2,20	19,75	20,19	0,08
	141e	2,70	20,40	20,97	1,25
	142e	2,90	19,35	19,93	-0,32
	143e	2,68	19,85	20,40	0,39
NUMBER OF RESULTS	-	50	50	50	-
OUTLIERS	**-**	**-**	**2**	**2**	**-**
AVERAGE	**-**	**2,44**	**19,65**	**20,14**	**-**
STD DEVIATION	**-**	**-**	**0,64**	**0,66**	**-**
MEDIAN	**-**	**-**	**19,77**	**20,19**	**-**
%RSD	**-**	**-**	**3,25**	**3,28**	**-**
ROBUST AVERAGE	**-**	**-**	**19,65**	**20,14**	**-**
ROBUST STD DEVIATION	**-**	**-**	**0,69**	**0,72**	**-**
UoM	**-**	**-**	**0,13**	**0,13**	**-**


ASH Z-SCORE
3,00
2,00
1,00
0,00
-1,00
-2,00
-3,00
Z-SCORE
8e
99e
139e
63e
122e
87e
1e
21e
59e
59e
62e
65e
33e
71e
79e
142e
127e
129e
29e
117e
3e
111e
5e
37e
140e
100e
4e
7e
10e
57e
143e
9e
134e
91e
128e
93e
53e
126e
89e
40e
12e
132e
113e
141e
66e
18e
11e
14e
25e*
24e*
LABORATORY ID

COAL CONCEPTS - PROFICIENCY TESTING - FEBRUARY 2026			
ANALYTICAL PARAMETER: ROGA INDEX			
	LAB ID	**ROGA**	**Z-SCORE**
	129e	**0**	**na**
Number of results	**-**	**1**	**-**
OUTLIERS	**-**	**-**	**-**

COAL CONCEPTS - PROFICIENCY TESTING - FEBRUARY 2026			
ANALYTICAL PARAMETER: FREE SWELLING INDEX			
	LAB ID	**FSI**	**Z-SCORE**
	1e	0,0	-
	37e	1.0	-
	91e	0,0	-
	111e	0,0	-
	127e	0.5	-
	129e	0.5	-
Number of results	**-**	**3**	
OUTLIERS	**-**	**0**	
Average		0,0	
Median		0,0	
Stdev		0,0	

3. CONCLUSIONS

3.1 The Total Moisture z-score trend has a balanced distribution. The Average, Robust average and Median are the same.

3.2 The Ash z-score trend shows a considerable spread around the mean. The Median and Robust Average are similar.

3.3 Homogeneity check:

SAMPLE NO.	TEST PORTION 1	TEST PORTION 2	sample av (Xt)	range (Wt)	range sqd
1	17,96	18,14	18,05	0,18	0,0324
2	17,94	18,10	18,02	0,16	0,0256
3	18,12	18,04	18,08	0,08	0,0064
4	18,14	17,98	18,06	0,16	0,0256
5	17,90	17,92	17,91	0,02	0,0004
6	18,08	17,89	17,99	0,19	0,0361
7	17,95	18,06	18,01	0,11	0,0121
8	18,00	18,10	18,05	0,10	0,0100
9	17,96	17,85	17,91	0,11	0,0121
10	17,92	17,84	17,88	0,08	0,0064
GENERAL AVERAGE			17,99		
STANDARD DEVIATION			0,072		
WITHIN SAMPLE STANDARD DEVIATION			0,091		
BETWEEN SAMPLE STANDARD DEVIATION			0,032		

The Total Moisture samples were confirmed to be sufficiently homogenous.

End of Report

COAL CONCEPTS: Terms and Conditions

Return of results:

Laboratories participate in proficiency testing programs on the understanding that they will be sharing their results and information **anonymously** with other laboratories performing the same analysis. No return of results compromises the spirit of the programs, and reports will not be sent to laboratories unless they return results. Payment in full is required from all laboratories enrolling whether they return results or not.

Errors in Participant Proficiency Testing Results:

Proficiency testing reports should reflect the level of accuracy that a regular testing client would receive.

If a participant finds an error in their proficiency testing results, they may notify us in writing and change their submission **PRIOR** to the due date for return. Changes after this time will not be accepted.

Coal Concepts' reports results *as submitted* by participants.

On occasion, it seems as though participants have mixed up the samples or not processed the samples according to the instructions. Coal Concepts cannot make assumptions of this nature and change results 'to suit'. We also cannot compromise the integrity of the programs by suggesting to some participants that they should review their results prior to the due date. (This is unfair to other participants) It is the responsibility of the participants to check all aspects of the program, including sample identification, preparation, testing instructions, calculations and reporting of the results prior to results submission.

If samples are not in good condition on arrival to the participant laboratory, Coal Concepts must be notified in writing IMMEDIATELY, as often samples can be replaced in good time. Claims about samples received in bad condition will not be accepted after the report has been issued.

Late Enrolments and Late Results:

Late enrolment requests cannot always be accommodated, as sample manufacture must be scheduled well in advance to the shipping date of the program to allow all necessary quality assurance activities to be carried out.

Shipping of PT materials and evaluating test results from PTPs out of cycle with the mainstream programs is considerably time consuming and therefore costly. In order not to disadvantage participants able to comply with time frames, Coal Concepts may charge a late fee in the following circumstances:

Requests that Coal concepts staff enters results on behalf of participants.

Requests to record results after the due date

Requests for PTP participation that is out of cycle with the scheduled dates

Shipping fees and Customs clearance:

Costs incurred for shipping samples and clearance of same through customs are the responsibility of the participating laboratory unless otherwise indicated

Non-payment of fees:

Coal Concepts retains the right to withhold reports and/or test materials and services when invoices are outstanding.

Confidentiality of results:

All data and information received by Coal Concepts from its clients are considered confidential unless the client has given express permission to pass on information.

Definitions:

The dictionary definitions of "collusion" and "falsification" are as follows.

· *Collusion:* A secret agreement or cooperation for a fraudulent or deceitful purpose.

· *Falsification:* Deliberately changing something to be false. In proficiency testing terms, collusion is comparing data (and perhaps changing data) to fit in with a believed "correct" result. This is contrary to the spirit of proficiency testing programs, which are issued with the intention of providing an objective comparison of a laboratory's performance with others. Coal Concepts tries to minimise the occurrence of collusion by being aware that laboratories should be objective when they report their results, and should therefore not know the intended results at the time they are reporting to us.

Answers are not provided to clients until results have been submitted.

To prevent collusion and falsification our advice to clients is:

DON'T confer with others about PT samples or results.

DO accept the fact that everyone makes errors.

DON'T average the results or opinions of every person in the laboratory before selecting the answer to be submitted. Instead, use one of the answers AS SUBMITTED to you and take advantage of the Coal Concepts internal QA services and submit all answers generated by the technicians.

DO have confidence in your own results.

Proficiency Testing (PT) is a compulsory part of laboratory accreditation, but it is also an important tool for giving you confidence in your results. "Enhancing" your PT results with assistance from another participant cannot increase confidence in your laboratory's performance.

Coal concepts' testing staff are not told what the expected results are, nor what we are expecting.

We subject ALL results to analysis, even if they are different.

The staff have the right to check that the results we enter on their behalf are correctly transcribed.

Clients are always welcome to contact Coal Concepts to seek advice or information about collusion or falsification of data.

Policy for Participant Appeal of PT Performance Assessment:

If participants disagree with their performance assessment in a proficiency report, they should inform Coal Concepts in writing.

The response will include Coal Concepts interpretation of the outcome of the reassessment and an explanation of that outcome. (For example, explanation of a calculation, or the rationale for the outcome of the evaluation.)

If a mistake has been made by Coal Concepts, it will be dealt with via Coal Concepts' non-conformance system.

Liability

In no event shall a party's liability to the other party for direct damages exceed an amount equal to the value of the amount for the PT Programme, under that specific month